Rakesh Dey

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Research Interest

I am passionate about comprehending the theoretical aspects of **Language Model**, with a specific focus on exploring interpretability and alignment. My previous research was mainly focused on unraveling the potential of the **latent structure** of **GAN**s & **Diffusion Model**s. In addition to this, I am also interested in exploring the **adversarial attack** in the **self-supervised contrastive learning** paradigm directing model robustness.

Education

University at Buffalo, SUNY — Buffalo, NY
Masters of Science in Computer Science and Engineering — *Aug 2026 – Present*

Maulana Abul Kalam Azad University of Technology (WBUT) — Kolkata, India
B.Tech. in Computer Science and Engineering; ***GPA: 8.77/10*** — *Aug 2017 – Jul 2021*

- **Relevant courses:** Artificial Intelligence (Introduction to Image Processing, Computer Vision), Mathematics I, II (Linear Algebra, Calculus, ODE, Probability, Statistics), Numerical Methods, Design & Analysis of Algorithms, Introductory Computer Graphics.

Experience

CVPR Unit, Indian Statistical Institute — Kolkata, India
Research Associate — *Oct 2022 – Feb 2026, Full-time*

- Worked on the advancement of self-supervised contrastive learning by dynamically varying temperature.
  Project: Dynamic temperature scaling for self-supervised contrastive learning
  Collaborator & supervisor: Siladittya Manna, ISI, Kolkata; Prof. Saumik Bhattacharya, IIT, Kharagpur; Prof. Umapada Pal, ISI, Kolkata
- Developed a novel method for estimating heart rate from facial video data.
  Project: Heart rate estimation from video
  Supervisor: Prof. Umapada Pal, ISI, Kolkata; Prof. Shivakumara Palaiahnakote, University of Salford, Manchester
- Created and implemented a system that utilizes facial video input to accurately predict real-time heart rates.
  Project: Heart rate estimation from video
  Supervisor: Prof. Umapada Pal, ISI, Kolkata; Prof. Shivakumara Palaiahnakote, University of Salford, Manchester

CVPR Unit, Indian Statistical Institute — Kolkata, India
Research Internship — *Apr 2022 – Oct 2022*

- Conducted a comprehensive survey on contemporary techniques for measuring blood oxygen saturation.
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata
- Created a dataset specifically designed for the estimation of heart rate from facial video data.
  Supervisor: Prof. Umapada Pal, CVPR Unit, Indian Statistical Institute, Kolkata

Tata Consultancy Service — Kolkata, India
Associate System Engineer Trainee — *Sept 2021 – Apr 2022*

- Participated in the application development and maintenance team for the business unit, contributing to software development and support activities

Publications

- **Rakesh Dey**, Shivakumara Palaiahnakote, Saumik Bhattacharya, Umapada Pal, Sukalpa Chanda, "A New StyleGAN Latent Space Based Model for Image Style Transfer" (*Accepted in* ***ICPR - 2024***)
- Siladittya Manna, Soumitri Chottopadhay, **Rakesh Dey**, Saumik Bhattacharya, Umapada Pal, "DySTreSS: Dynamically Scaled Temperature in Self-Supervised Contrastive Learning" (*Accepted in* ***IEEE Transactions on Artificial Intelligence***, arxiv)
- **Rakesh Dey**, Shivakumara Palaiahnakote, Saumik Bhattacharya, Umapada Pal, Sukalpa Chanda, "SFTT: A Spatial-Frequency-Temporal based End-to-End Transformer for Heart Rate Estimation" (*Accepted in* ***IEEE Transactions on Emerging Topics in Computational Intelligence***)

- Arpan Bairagi, **Rakesh Dey**, Siladittya Manna, Umapada Pal, "Reliability-Aware Weighted Multi-Scale Spatio-Temporal Maps for Heart Rate Monitoring" (*Accepted in **ICIP - 2026***)
- **Rakesh Dey**, "Can Entropy Regularization Prevent Dimension Collapse in Spectral Contrastive Learning?" (*Under review*)
- **Rakesh Dey**, Subhajit Maity, "CariSketcher: Generate Identity Preserved Caricature Guided by Sketch" (*Under review*)
- **Rakesh Dey**, Shivakumara Palaiahnakote, Umapada Pal, Sukalpa Chanda, and Yue Lu, "Blood Oxygen Saturation Measurement: A Survey" (*Accepted in **Artificial Intelligence and Applications***)

Research Projects

Aligning Video Diffusion Models with Text Prompt Kolkata, India
Collaborators: Prof. Umapada Pal, Prof. Shivakumara Palaiahnakote *Jun 2025 – Jan 2026*
Investigated text-conditioning strategies for improving prompt–video semantic alignment in diffusion-based video generation models.

Dynamic Temperature Scaling in Self-supervised Contrastive Learning Kolkata, India
Collaborators: S. Manna, Prof. S. Bhattacharya, Prof. U. Pal *Mar 2023 – Nov 2023*
Studied the impact of feature cosine similarity on temperature hyperparameter to enhance uniformity and alignment for better feature distribution in embedding space in self-supervised contrastive learning.

Sketch-guided Caricature Generation using StyleGAN Kolkata, India
Collaborators: Subhajit Maity, University of Central Florida, USA *Feb 2024 – Sep 2024*
Investigated the influence of sparse line drawings, sketches and edge maps for image manipulation with StyleGAN.

Exploring StyleGAN Latent Space for Image Editing Task Kolkata, India
Collaborators: Prof. S. Palaiahnakote, Prof. S. Bhattacharya, Prof. S. Chanda, Prof. U. Pal *Oct 2023 – Apr 2024*
Explored optimization-based inversion techniques on latent space of StyleGAN for performing image editing tasks.

Heart Rate Estimation from Facial Video Kolkata, India
Collaborators: Prof. Shivakumara Palaiahnakote, Prof. Umapada Pal *Dec 2022 – Oct 2023, May 2025 – Dec 2025*
Examined the impact of spatial and frequency domain information on heart rate estimation.

Survey on Blood Oxygen Saturation Measurement Kolkata, India
Collaborators: Prof. Shivakumara Palaiahnakote, Prof. Umapada Pal *Jan 2022 – Jun 2022*
Conducted a comprehensive study on blood oxygen saturation measurement methods and explored the potential of computer vision techniques for the same.

Technical Skills

Programming Languages: Python, C/C++

Libraries: Pytorch, OpenCV, Scikit-Learn, Keras, TensorFlow, NumPy, Pandas, Matplotlib, dlib

Documentation & Typesetting: LaTeX, MS Office

Technologies: Git, Linux

Awards, Achievements & Certificates

Scholarship: "Half Tuition Fee Waiver for Excellent Academic Records" for four consecutive years (2017 - 2021).
Achievements: Secured 2nd & 1st position in college & school art competitions respectively.
Certificates: Certificate of excellence for an outstanding summer project from GLOBSYN BUSINESS SCHOOL (2018).

Volunteer Services

- Peer reviewer: **Springer Nature Computer Science**, **ICASSP 2025**
- Volunteering experience: International Conference on Pattern Recognition **(ICPR)** 2024
- Invited talk/demo presentation: **Deep Learning for Physiological Measurements**
  Gave a talk on Deep Learning techniques for estimating physiological parameters and presented a demo for estimating heart rate from facial video.